# Community-Acquired Pneumonia (CAP): Rapid Evidence Review 2025–2026

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**Sources:** ATS 2025 Guidelines, IDSA/ATS 2019, UCSF IDMP, AAFP 2022

## Abstract

Community-acquired pneumonia (CAP) remains a leading cause of infectious disease mortality worldwide, with an annual incidence of 248 per 100,000 adults. This review synthesizes the latest 2025 ATS guidelines with established IDSA/ATS 2019 recommendations, providing updated evidence on diagnosis, empiric antibiotic therapy, duration of treatment, and prevention strategies.

## 1. Epidemiology

- **Annual incidence:** 248 cases per 100,000 adults

- **Incidence increases dramatically with age:**
  - Ages 65–79: 634 per 100,000
  - Ages ≥80: 16,430 per 100,000
- **30-day mortality:** 6% for hospitalized patients; 34% for those with treatment failure
- **Hospitalization rate:** 9× higher in COPD patients
- **Microbiology (hospitalized CAP):**
  - No pathogen identified: 62%
  - Virus only: 23%
  - Bacteria only: 11%
  - Mixed bacteria + virus: 13%

**Common bacterial pathogens:** | Pathogen | Frequency | |———-|———–| | *Streptococcus pneumoniae* | Most common | | *Haemophilus influenzae* | Common | | *Mycoplasma pneumoniae* | Common (atypical) | | *Chlamydia pneumoniae* | Common (atypical) | | *Legionella* spp. | 3% of hospitalized | | *Moraxella catarrhalis* | Less common | | Respiratory viruses | 23% |

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## 2. Diagnosis

### Clinical Presentation

- **Classic symptoms:** Cough, fever, chills, sputum production, dyspnea, pleuritic chest pain
- **Signs:** Tachypnea, tachycardia, hypoxia, altered mental status (especially elderly)
- **High diagnostic odds ratios:**
  - Physician's overall clinical impression (OR = 11.5)
  - Egophony (OR = 6.5)

- Any abnormal vital sign (OR = 6.0)
- Tachypnea (OR = 3.1)
- Measured fever (OR = 3.3)

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> ⚠️ **Elderly patients may NOT present with fever**

### Imaging

- **Chest X-ray (CXR):** Standard for radiographic confirmation
  - PPV only 26.9% vs CT (NPV 96.5%)
- **CT chest:** More accurate but cost/availability limited
- **Lung Ultrasound (LUS):** Now acceptable alternative when CXR/CT not readily available (ATS 2025)
  - More accurate than CXR per meta-analysis
  - Requires trained operator

## Laboratory Testing

| Test | Recommendation |
|---|---|
| CBC | Often elevated WBC |
| Sputum culture | NOT routine; for severe CAP, MRSA/Pseudomonas risk factors |
| Urine antigen tests | NOT routine; for severe CAP |
| Procalcitonin | NOT recommended (IDSA) |
| Influenza/COVID testing | If flu-like symptoms during high-activity season |
| MRSA nares | If empiric vancomycin initiated (high negative predictive value) |

## 3. Severity Assessment

### CURB-65 Score

| Criterion | Points |
|---|---|
| **C**onfusion (new onset) | 1 |
| **U**rea > 7 mmol/L | 1 |
| **R**espiratory rate ≥ 30/min | 1 |
| **B**P < 90/60 mmHg | 1 |
| Age ≥ **65** years | 1 |

| Score | Risk | Action |
|---|---|---|
| 0–1 | Low (mortality 0.5–5%) | Outpatient |
| 2 | Moderate (mortality 5.1%) | Consider hospitalization |
| 3–4 | High (mortality 18.9%) | Hospitalize; consider ICU |

### PSI/PORT Score

- Validated 20-variable score
- Use as adjunct to clinical judgment (IDSA recommendation)
- Available at: mdcalc.com/psi-port-score-pneumonia-severity-index-cap

## 4. Treatment

### Outpatient Empiric Therapy

#### Previously Healthy, No Antibiotics in Past 3 Months

| Option | Dose | Notes |
|---|---|---|
| **Doxycycline** ⭐ | 100 mg PO BID | **Preferred** |
| Amoxicillin | 1 g PO TID | Alternative |
| Macrolide* | Standard dose | Only if local macrolide resistance <25% |

*Macrolide options: Azithromycin, Clarithromycin

#### Comorbidities OR Antibiotic Use in Past 3 Months

**Comorbidities:** Chronic heart/lung/liver/renal disease, DM, alcoholism, malignancy, asplenia, immunosuppression

| Regimen | Dose | Notes |
| --- | --- | --- |
| **Doxycycline + Amoxicillin** ⭐ | 100 mg BID + 1 g TID | **Preferred combination** |
| Doxycycline + Cefpodoxime | 100 mg BID + 200 mg BID | Alternative combination |
| **Levofloxacin** (monotherapy) | 750 mg PO daily | Alternative; contact ASP if requiring approval |

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⚠️ **ATS 2025 Update:** For outpatients WITH comorbidities who test positive for respiratory virus (influenza/COVID): consider NOT adding empiric antibiotics if clinically stable — to reduce unnecessary antibiotic use.

## Inpatient Empiric Therapy

### Non-Severe CAP

- **Beta-lactam + Macrolide** OR
- **Beta-lactam + Doxycycline**

**Beta-lactam options:** Ceftriaxone, Cefotaxime, Ampicillin-sulbactam

**Macrolide:** Azithromycin (500 mg daily × 3 doses for atypicals)

### Severe CAP

- **Combination therapy required:**
  - Beta-lactam + Macrolide OR
  - Beta-lactam + Doxycycline

**If MRSA risk factors (prior isolation, recent hospitalization + parenteral antibiotics):** - Add: Vancomycin OR Linezolid

**If Pseudomonas risk factors:** - Add: Piperacillin-tazobactam, Cefepime, Ceftazidime, Imipenem, Meropenem, OR Aztreonam

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> 💡 **Rule:** Obtain cultures BEFORE escalating coverage to allow de-escalation if pathogen not confirmed

### Special Populations

**Influenza with CAP:** - Add Oseltamivir 75 mg PO BID × 5 days - Reduce dose in renal insufficiency - ICU/immunocompromised/obese patients may need higher doses/prolonged therapy

## 5. Duration of Antibiotic Therapy

### ATS 2025 Key Change — Shortened Duration

| Setting | Recommended Duration |
|---|---|
| Outpatient (non-severe, stable) | **3–5 days** |
| Inpatient non-severe CAP (stable) | **3–5 days** |
| Confirmed MRSA | ≥7 days |
| Confirmed Pseudomonas | ≥7 days |
| Legionella suspected | 7 days |
| Typical uncomplicated | Minimum 5 days |

**Stop antibiotics when:** - Afebrile 48–72 hours - Stable (HR ≤100, RR ≤24, SBP ≥90, $O_2$ sat ≥90%, alert)

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> ⚠️ CXR and cough abnormalities may persist **up to 6 weeks** — NOT a valid reason to extend antibiotics

## 6. Corticosteroids in CAP

| Setting | Recommendation |
| --- | --- |
| Non-severe CAP (inpatient) | **NOT recommended** ❌ |
| Severe CAP | **Suggested** ✅ (conditional) |
| Refractory septic shock | May be considered |

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Evidence: May reduce ARDS risk in severe CAP but **no mortality benefit** overall

## 7. Prevention

### Pneumococcal Vaccination (2022 Updated Guidelines)

| Population | Vaccine Recommendation |
|---|---|
| Adults ≥65 years | **PCV20 alone** OR **PCV15 + PPSV23** (1 year later) |
| Ages 19–64 with comorbidities | **PCV20 alone** OR **PCV15 + PPSV23** (1 year later) |
| Previously received PCV13 only | No additional PCV20/PCV15 needed |
| Previously received PPSV23 only | Give PCV15 or PCV20 **1 year after PPSV23** |

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PCV13 no longer routinely recommended for immunocompetent adults ≥65 years

**PPSV23 efficacy:** 13% relative risk reduction in CAP; 28% reduction in adults ≥65 or high-risk younger adults

### Other Vaccinations

- **Influenza:** Annual vaccination recommended for all adults
- **COVID-19:** Per current CDC guidelines

## 8. Key Updates: ATS 2025 vs. IDSA/ATS 2019

| Topic | 2019 IDSA/ATS | 2025 ATS Update |
|---|---|---|
| **Antibiotic duration** | ≥5 days | **3–5 days** if clinical stability achieved |
| **Lung ultrasound** | Not specified | Acceptable alternative to CXR |
| **Corticosteroids (non-severe)** | Not recommended | Explicitly **against** |
| **Corticosteroids (severe)** | Not recommended | **Suggested** (conditional) |
| **Viral-positive outpatient** | Antibiotics always | Consider NO antibiotics if stable (with comorbidities) |
| **Outpatient doxycycline** | Alternative | **Preferred** first-line |

## 9. Take-Home Points

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1. **Diagnose CAP** with clinical features + radiographic infiltrate (CXR, CT, or LUS)
2. **Assess severity** with CURB-65 to guide outpatient vs. inpatient decision

3. **Start empiric antibiotics** promptly: Doxycycline for most outpatients; combination therapy for comorbidities
4. **Shorten duration** to 3–5 days if clinical stability achieved (ATS 2025)
5. **De-escalate** when cultures return; stop broad-spectrum coverage if MRSA/Pseudomonas not confirmed
6. **No corticosteroids** for non-severe CAP
7. **Prevent** with PCV20/PCV15 + PPSV23 vaccination in appropriate populations

## References


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